

Mathematical Foundations of Optical Interferometry: Embedding Arbitrary Matrices into Mach-Zehnder Interferometers

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1 Introduction

This section provides a rigorous mathematical derivation of beam splitters, mirrors, Mach-Zehnder interferometers (MZIs), and their use in performing matrix multiplications. We also present the complex Schur decomposition and its role in embedding an arbitrary matrix into an MZI-based optical computing architecture.

2 Mathematical Description of Optical Components

2.1 Beam Splitters

A 50:50 beam splitter transforms the input field amplitudes as follows:

$$BS = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}. \quad (1)$$

This matrix ensures energy conservation and unitary behavior.

2.1.1 Proof of Beam Splitter Matrix

To show that the beam splitter matrix is unitary, we compute:

$$BS^H BS = \frac{1}{2} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} = I. \quad (2)$$

This confirms that the beam splitter transformation preserves norm and is unitary.

2.2 Derivation of the Beam Splitter Matrix

A **50:50 beam splitter** takes an incident optical field and **splits it into two paths** while introducing a phase shift in the reflected component.

2.2.1 Define the Input and Output Fields

Let the input complex field amplitudes for the two ports be:

$$\begin{bmatrix} E_1^{\text{in}} \\ E_2^{\text{in}} \end{bmatrix} \quad (3)$$

where: - E_1^{in} represents the field entering from port 1. - E_2^{in} represents the field entering from port 2.

The output fields $E_1^{\text{out}}, E_2^{\text{out}}$ are related to the input by a transformation matrix BS :

$$\begin{bmatrix} E_1^{\text{out}} \\ E_2^{\text{out}} \end{bmatrix} = BS \begin{bmatrix} E_1^{\text{in}} \\ E_2^{\text{in}} \end{bmatrix} \quad (4)$$

2.2.2 Form of the Beam Splitter Matrix

By experimental observation and symmetry considerations, the transformation must be **unitary** and **preserve energy**. We assume the general form:

$$BS = \begin{bmatrix} t & r \\ r & t \end{bmatrix} \quad (5)$$

where: - t is the **transmission coefficient**. - r is the **reflection coefficient**.

2.2.3 Enforcing Energy Conservation

Energy conservation in an optical system requires that the total intensity (power) remains unchanged:

$$|E_1^{\text{out}}|^2 + |E_2^{\text{out}}|^2 = |E_1^{\text{in}}|^2 + |E_2^{\text{in}}|^2. \quad (6)$$

This implies that BS must be **unitary**, meaning:

$$BS^\dagger BS = I. \quad (7)$$

Expanding,

$$\begin{bmatrix} t^* & r^* \\ r^* & t^* \end{bmatrix} \begin{bmatrix} t & r \\ r & t \end{bmatrix} = \begin{bmatrix} |t|^2 + |r|^2 & t^*r + r^*t \\ r^*t + t^*r & |t|^2 + |r|^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (8)$$

From this, we obtain two conditions:

1. Energy Conservation Condition:

$$|t|^2 + |r|^2 = 1. \quad (9)$$

2. Cross-Term Cancellation:

$$t^*r + r^*t = 0. \quad (10)$$

Since t and r are complex numbers, we let:

$$t = \frac{1}{\sqrt{2}}, \quad r = \frac{i}{\sqrt{2}}. \quad (11)$$

which satisfies the unitary condition:

$$\left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{i}{\sqrt{2}}\right)^2 = \frac{1}{2} + \frac{1}{2} = 1. \quad (12)$$

Thus, the **beam splitter matrix** is:

$$BS = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}. \quad (13)$$

2.2.4 Verification of Unitarity

To verify that this matrix is **unitary**, we compute:

$$BS^\dagger BS = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}. \quad (14)$$

Expanding:

$$\frac{1}{2} \begin{bmatrix} 1+1 & 0 \\ 0 & 1+1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \quad (15)$$

Thus, the matrix is **unitary**, completing the proof.

2.3 Mirrors

Mirrors introduce a phase shift upon reflection:

$$M = \begin{bmatrix} e^{i\phi} & 0 \\ 0 & e^{i\psi} \end{bmatrix}, \quad (16)$$

where ϕ and ψ are phase shifts depending on the optical path.

3 Mach-Zehnder Interferometer (MZI)

An MZI consists of two beam splitters and phase shifters, modeled as:

$$U_{\text{MZI}} = BSPBS, \quad (17)$$

where P is a diagonal phase shift matrix:

$$P = \begin{bmatrix} e^{i\theta_1} & 0 \\ 0 & e^{i\theta_2} \end{bmatrix}. \quad (18)$$

The resulting transformation matrix remains unitary.

4 Complex Schur Decomposition

For any arbitrary matrix $A \in \mathbb{C}^{n \times n}$, there exists a unitary decomposition:

$$A = QTQ^H, \quad (19)$$

where:

- Q is a unitary matrix (can be implemented with fixed beam splitters),
- T is an upper triangular matrix (implemented using phase shifters).

5 Embedding an Arbitrary Matrix into an MZI for Matrix Multiplication

Given an arbitrary 2×2 matrix A , we compute the Schur decomposition:

$$A = QTQ^H. \quad (20)$$

5.1 Implementing Q Using Fixed Beam Splitters

Since MZIs implement unitary matrices of the form:

$$BS = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}, \quad (21)$$

we can directly use an MZI network to implement Q and Q^H .

5.2 Implementing T Using Phase Shifters

Since T is upper triangular with eigenvalues on the diagonal:

$$T = \begin{bmatrix} e^{i\theta_1} & t_{12} \\ 0 & e^{i\theta_2} \end{bmatrix}, \quad (22)$$

this can be implemented by setting appropriate phase shifts within the MZI.

5.3 Final Optical Implementation

1. Apply Q^H using a fixed MZI beam splitter network.
2. Apply T using phase shifters.
3. Apply Q using a second MZI beam splitter network.

Thus, Ax is implemented optically as:

$$Ax = QTQ^Hx. \quad (23)$$